MODEL NAME :	QWEN 3.8	BONSAI27B
PUBLISHER:	ALIBABA CLOUD	**PrismML**.
PARAMETER:	27B	27B
LAUNCH DATE :	15TH AUGUST	10TH AUGUST
ARCH :	QWEN FAMILY(27B)	QWEN 3.6 (27B)
MODEL TYPE :	Qwen3.8-27B is a dense 27B vision-language model for coding, professional work, research, and long-horizon agentic tasks, with configurable reasoning and a native 262K-token context window.	1-bit model with mission reasoning and tool use
TOKEN CONTEXT WINDOW	262,000	262,000
MODEL ARCHITECTURE	Dense model with qwen 3.5 architecture .	1-bit binary model/ternary model of qwen 3.6 architecture.
LM Output:	248,320 (Padded)	
MTP (Multi-Token Prediction):	trained with multiple steps	
Number of Linear Attention Heads:	48 for V and 16 for QK	

Feature	Binary	Ternary
Base Model	Qwen3.6-27B hybrid model	Qwen3.6-27B hybrid model
Parameters	~27.3B language + 0.46B vision	~27.3B language + 0.46B vision
Architecture	Hybrid attention, SwiGLU, RoPE, RMSNorm	Hybrid attention, SwiGLU, RoPE, RMSNorm
Context Length	262K tokens	262K tokens
KV Cache	4-bit; ~4.3 GB	4-bit; ~4.3 GB

Weight Format	1-bit: {-1,+1} + FP16 scaling	2-bit: {-1,0,+1} + FP16 scaling
Low-Bit Coverage	Embeddings, attention, MLP, LM head	Embeddings, attention, MLP, LM head
Vision Tower	4-bit HQ; optional 0.63 GB	4-bit HQ; optional 0.63 GB
Deployed Size	~3.9 GB	~7.2 GB
Acceleration	DSpark speculative decoding	DSpark speculative decoding

TOKENISATION OF DATASET : AIR QUALITY

DATASET URL : https://archive-beta.ics.uci.edu/dataset/360/air%2Bquality/files

FEATURES	CONTENTS
Type of Dataset	NUMERICAL
Number of samples	9358
Number of features	15
Dataset context	Contains the responses of a gas multisensor device deployed on the field in an Italian city. Hourly responses averages are recorded along with gas concentrations references from a certified analyzer. The dataset contains 9358 instances of hourly averaged responses from an array of 5 metal oxide chemical sensors embedded in an Air Quality Chemical Multisensor Device. The device was located on the field in a significantly polluted area, at road level,within an Italian city. Data were recorded from March 2004 to February 2005 (one year)representing the longest freely available recordings of on field deployed air quality chemical sensor devices responses. Ground Truth hourly averaged concentrations for CO, Non Metanic Hydrocarbons, Benzene, Total Nitrogen Oxides (NOx) and Nitrogen Dioxide (NO2) and were provided by a co-located reference certified analyzer.

	Evidences of cross-sensitivities as well as both concept and sensor drifts are present as described in De Vito et al., Sens. And Act. B, Vol. 129,2,2008 (citation required) eventually affecting sensors concentration estimation capabilities. Missing values are tagged with -200 value. This dataset can be used exclusively for research purposes. Commercial purposes are fully excluded.
DOMAIN OF THE DATSET	Computer Science (Environment focused)
EXISTING PRODUCTS	1. Google Maps Platform – Air Quality AP https://developers.google.com/maps/documentation/air-quality?utm_source=chatgpt.com 2. IBM Environmental Intelligence Suite .https://www.ibm.com/products/environmental-intelligence-suite?utm_source=chatgpt.com 3. Kunak AIR Cloud https://kunakair.com/air-quality-software/?utm_source=chatgpt.com 4.IQAir AirVisual Platform https://www.iqair.com/in-en/commercial-air-quality-monitors/airvisual-platform?utm_source=chatgpt.com 5.PurpleAir – Real-Time Air Quality Monitoring https://www.purpleair.com/
OPEN SOURCE PRODUCTS	1. **SWELTER** - https://github.com/ChelseaKR/swelter 2. **OpenAQ Explorer** - https://github.com/openaq/openaq-explorer 3. **OpenAQ Python SDK** - https://github.com/openaq/opena

	q-python 4. **Ozon3** - https://github.com/Ozon3Org/Ozon3 5. **MySense** - https://github.com/teusH/MySense 6. **PLUME Dashboard** - https://github.com/atmosud/plume-dashboard

EXISTING PRODUCT FEATURES :

No.	Feature / Aim	Google Maps Air Quality API	IBM Environmental Intelligence Suite	Kunak AIR Cloud	IQAir AirVisual Platform	PurpleAir
1	**Real-Time Monitoring**	Current air-quality conditions	Real-time environmental monitoring	Real-time air-quality monitoring	Real-time AQ monitoring	Real-time sensor readings
2	**AQI Monitoring**	AQI and pollutant information	Air-quality indicators & environmental metrics	AQI calculations	AQI and air-quality levels	AQI visualization
3	**Pollutant Monitoring**	Multiple pollutants including CO, NO_2, O_3, PM2.5	Air-quality & environmental parameters	Multiple pollutants and gases	PM2.5, PM10, O_3, NO_2 and others	PM1, PM2.5, PM10 and other measurements
4	**Historical Data**	Hourly historical air-quality data	Historical environmental data analysis	Historical sensor data	Historical AQ data	Historical sensor data
5	**Map Visualization**	Air-quality heatmaps & location-based data	Geospatial maps	Interactive air-quality maps	Global AQ map	Live global sensor map

6	**Data Analytics**	Air-quality data analysis	Advanced environmental analytics	Sensor & pollution analytics	AQ trends and analysis	Sensor data analysis & trends
7	**Forecasting**	Up to 96-hour air-quality forecasts	Environmental forecasting	—	Air-quality forecasts	—
8	**Alerts / Notifications**	—	Environmental alerts	Pollution & threshold alerts	AQ alerts	Alerts / notifications
9	**Sensor Monitoring**	—	IoT/sensor data integration	Sensor/device monitoring	AirVisual sensor management	Connected sensor monitoring
10	**API / Data Integration**	Air Quality API	Data & IoT integration	API & data integration	Platform/API integration	API access to sensor data

OPEN SOURCE PRODUCT FEATURES :

No.	Feature / Aim	SWELTER	OpenAQ Explorer	Ozon3	MySense	PLUME Dashboard
1	**Real-Time Monitoring**	Live/community sensor observations and model data	Displays air-quality monitoring data	Retrieves current AQ data	Collects sensor measurements continuously	Real-time mobile air-pollution monitoring
2	**AQI Monitoring**	AQ-related data/QC and visualization	AQ data exploration and visualization	AQI readings from supported AQ sources	Displays measured environmental values; AQI depends on implementation	Air-pollution levels and threshold-based monitoring
3	**Pollutant Monitorin**	Air-quality pollutants	Multiple pollutant	Pollutant/AQI data	PM, gases and	Mobile air-pollutio

	g	from supported sources	measurements from OpenAQ	through AQ APIs	environmental sensor measurements	n measurements
4	**Historical Data**	Stores/version-controls observations and snapshots	Explore historical OpenAQ measurements	Historical AQ data retrieval	Stores/forwards measurements for databases	Historical/post-processing of collected measurements
5	**Map Visualization**	Live map/data visualization	Map-based air-quality exploration	Data can be consumed by applications	Location/GPS-supported measurements and visualization	GPS-based pollution mapping
6	**Data Analytics**	Validation, QC, calibration corrections and uncertainty	Data exploration and filtering	AQ-data processing/retrieval	Sensor data processing and calibration	Peak detection, baseline subtraction and post-processing
7	**Forecasting**	Integrates modelled/reanalysis air-quality sources	Primarily data exploration rather than forecasting	Provides forecasted AQ data through supported sources	Forecasting is described as a planned/possible capability rather than a core mature feature	Not primarily a forecasting platform
8	**Alerts / Notifications**	Alert/webhook mechanisms documented	Not a primary feature	Not a primary feature	Warning/notice system	Threshold-based alerts
9	**Sensor Monitoring**	Sensor observations, QC, calibration provenance	Displays monitoring-station data	Depends on external AQ data sources	**Core feature** — sensor collection, device status and remote	Mobile sensor/measurement workflow

					managem ent	
10	**API / Data Integratio n**	OpenAQ, Sensor.Co mmunity, Open-Met eo and interopera bility workflows	OpenAQ data/API ecosystem	AQ APIs and Python integration	MQTT, databases , Influx, JSON and other data channels	Data import/exp ort and mobile monitoring workflow

COMMON FEATURES

No.	Common Feature	Existing Products	Open-Source Products	Why it is common
1	**Real-Time Air-Quality Monitoring**	Google, IBM, Kunak, IQAir, PurpleAir	SWELTER, OpenAQ Explorer, Ozon3, MySense, PLUME	Both groups support accessing/displaying current or continuously collected air-quality measurements
2	**Pollutant Monitoring**	Google, IBM, Kunak, IQAir, PurpleAir	SWELTER, OpenAQ Explorer, Ozon3, MySense, PLUME	Both work with measurements of pollutants such as PM, CO, NO_2, O_3, etc.
3	**Historical Data**	Google, IBM, Kunak, IQAir, PurpleAir	SWELTER, OpenAQ Explorer, Ozon3, MySense, PLUME	Both allow previous measurements to be stored, retrieved, explored, or analyzed
4	**Map / Location-Based Visualization**	Google, IBM, Kunak, IQAir, PurpleAir	SWELTER, OpenAQ Explorer, MySense, PLUME	Both use geographic location to display monitoring stations/readings or pollution information
5	**Data Analytics / Trend Analysis**	Google, IBM, Kunak, IQAir, PurpleAir	SWELTER, OpenAQ Explorer, Ozon3, MySense, PLUME	Both provide some form of processing, analysis, trends, statistics, QC, or visualization of AQ data

6	**Sensor / Monitoring-Station Management**	IBM, Kunak, IQAir, PurpleAir	SWELTER, OpenAQ Explorer, MySense, PLUME	Both ecosystems work with physical sensors/monitoring stations and their measurements
7	**API / Data Integration**	Google, IBM, Kunak, IQAir, PurpleAir	SWELTER, OpenAQ Explorer, Ozon3, MySense	Both can connect to external data sources, APIs, databases, or data pipelines
8	**AQI / Air-Quality Level Representation**	Google, IBM, Kunak, IQAir, PurpleAir	Ozon3 and related AQ-data workflows; others primarily expose pollutant measurements rather than a universal AQI	Both groups can represent air-quality condition/level information, although the implementation differs
9	**Alerts / Threshold Monitoring**	IBM, Kunak, IQAir, PurpleAir	SWELTER, PLUME, MySense	Both have projects/products capable of triggering or supporting alerts based on environmental conditions or thresholds
10	**Forecasting / Predictive Information**	Google, IBM, IQAir	Ozon3 and some model/data-source workflows	Present in both groups, but **not a universal feature** across all products